

[YIJC/2025/Q3]

- 3 A beginner archer practises his shots on a 9 x 9 target board. The board is laid out as a grid with rows and columns numbered from 1 to 9. The bullseye is located at the centre of the board, at coordinates (5, 5) .

	1	2	3	4	5	6	7	8	9
1	•	•	•	•	•	•	•	•	•
2	•	•	•	•	•	•	•	•	•
3	•	•	•	•	•	•	•	•	•
4	•	•	•	•	•	•	•	•	•
5	•	•	•	•	O	•	•	•	•
6	•	•	•	•	•	•	•	•	•
7	•	•	•	•	•	•	•	•	•
8	•	•	•	•	•	•	•	•	•
9	•	•	•	•	•	•	•	•	•

Bullseye at (5,5)

A target board is displayed with 9 rows of 9 dots, with labels 1 to 9 at the top and left side of the grid. The top left corner dot has $x=1$ and $y=1$. The bullseye at position (5, 5) is marked with the uppercase letter 'O'.

Task 3.1

Write program code to initialize a 2D array to represent the grid.

Write program code for the function `print_grid()` to display the labels, the dots, and the bullseye 'O' using the 2D array grid created. [4]

The archer begins with an initial skill level represented by a variable d , which limits how far the arrow can land from where the archer aims. The archer begins with an initial skill level of $d=5$, and he chooses where to aim by entering the x - and y -coordinates (x_1 , y_1).

The simulation program generates the actual strike point of the arrow (x_2 , y_2) where:

- x_2 is a random integer between (x_1-d) and (x_1+d) (inclusive)
- y_2 is a random integer between (y_1-d) and (y_1+d) (inclusive).

Depending on the outcome, the output can be one of the following:

If the strike point (x_2 , y_2) is outside the grid (i.e. not in the range 1 to 9 for both x and y), an output message 'You have missed the target board. Try again.' is printed.

- If the strike point is within the grid, the grid is printed with the hit coordinates (x_2 , y_2) marked with an 'X', if it is not the bullseye 'O'
- If the strike point is on the bullseye 'O', the grid is printed with the bullseye 'O' replaced with a 'H'.

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Task 3.2

Write program code to:

- prompt the archer to enter the x- and y-coordinates as his aiming coordinates
- generate the strike point (x_2 , y_2)
- if the strike point is within the grid, display the grid marking the strike point with an 'X'
- if the strike point is outside the grid, display the appropriate message. [5]

To compute the distance between (x_1 , y_1) and (x_2 , y_2), the following formula can be used.

$$\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Task 3.3

Write program code to:

- compute the distance between (x_1 , y_1) and (x_2 , y_2) using the formula above
- if the distance is less than d , update d to this smaller value to simulate the archer improving his aiming precision. [2]

Task 3.4

Write program code to simulate the beginner archer practising with the simulation program until he hits the bullseye.

Your program code should include the following:

- print a new target grid at the beginning of the training
- prompt the archer to enter the aiming coordinates or choose to quit the training by entering (0, 0) as his aiming coordinates
- generate the coordinates of the strike point
- print an updated grid with the strike point marked with 'X' if it is within the grid
- calculate the distance between the aiming coordinates and the strike point and update the value of d , if necessary
- repeat prompting the archer to enter new aiming coordinates until he hits the bullseye or when he chooses to quit the training
- display the number of attempts made by the archer when he hits the bullseye or chooses to quit. [12]